

Tan Xin Er

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience developing web, mobile, and IoT applications. Experienced in building AI-powered applications, automation tools, and intelligent systems. Passionate about full-stack development, scalable systems, and leveraging artificial intelligence to solve real-world problems.

EDUCATION

Bachelor of Engineering in Information & Communications Technology
majoring in Software Engineering (Honours with Merit) | May 2026

Singapore Institute of Technology (SIT), Singapore

Diploma in Accountancy | April 2022
Singapore Polytechnic (SP), Singapore

WORK EXPERIENCE

IT Application Intern

SBS Transit Pte Ltd

Singapore

May 2025 – April 2026

- Developed an AI-powered technical document generator that analyzes .NET modules and SQL scripts. Leveraged static code analysis and automated diagramming to provide instant insights into application architecture, logic flows, data interactions, and user purpose.
- Designed and implemented backend features for an application using ASP.NET (C#). Created and maintained SQL stored procedures for new integrations, features, and workflow optimizations, ensuring strict data integrity.

Data Engineer Intern

Munich Re

Singapore

December 2024 – April 2025

- Developed R scripts to perform ETL (Extract, Transform, Load) processes, automating data pipelines for efficient data integration and workflow optimization.

Junior Research Assistant Intern

Illinois Advanced Research Centre

Singapore

May 2024 – July 2024

- Engineered a web-based network monitoring dashboard to analyze cybersecurity risks across a complex IoT and ICS ecosystem, integrating data from robotics, microgrids, and building management systems.

PROJECTS

Secure Backend System with AWS IAM Authentication

- Developed and tested a RESTful API to support backend data retrieval and business workflows, implementing SQL-based data handling and validation logic.
- Secured API endpoints using AWS IAM (users, roles, and policies) and built a C# console application to consume and validate authenticated endpoints end-to-end.

Real-Time Multi-Modal Assistive Intelligence System

- Developed a multi-modal AI system for real-time sign language translation, speech-to-text, and facial emotion recognition using Raspberry Pi edge devices, integrating computer vision and speech recognition models with a React-based dashboard for real-time interaction.
- Designed and implemented a distributed, low-latency architecture using MQTT and Socket.IO to transmit sensor data and AI inference results between edge devices and backend services, enabling real-time processing and visualization.

SKILLS

JavaScript, TypeScript, React, Node.js, REST APIs, SQL, AWS, Git, Python, C#, Flutter

HACKATHONS

- Designed and implemented a custom LED control system on an ESP8266 microcontroller, integrating hardware wiring, controller logic, and MQTT-based communication for real-time remote brightness monitoring and adjustment, improving display visibility and energy efficiency.

CERTIFICATIONS

- SCRUM Master
- Microsoft Office Excel & Excel Expert

LANGUAGES

English: Fluent

Chinese: Intermediate

Bahasa Indonesia: Basic